

Nielsen & Chuang, Chapter 2 Exercises

The Pauli matrices are

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Exercise 2.11: Eigendecomposition of the Pauli matrices

Find the eigenvectors, eigenvalues, and diagonal representations of the Pauli matrices X , Y , and Z .

An eigenvector of A with eigenvalue λ is a non-zero vector $|v\rangle$ with $A|v\rangle = \lambda|v\rangle$. The eigenvalues are the solutions of the characteristic equation $c(\lambda) = \det[A - \lambda I] = 0$. For each eigenvalue λ , the eigenvectors are found by solving $(A - \lambda I)|v\rangle = 0$ for $|v\rangle = (a, b)^T$, and the solution is normalized so that $\langle v|v\rangle = |a|^2 + |b|^2 = 1$. A diagonal representation has the form $A = \sum_i \lambda_i |i\rangle\langle i|$, where the vectors $|i\rangle$ form an orthonormal set of eigenvectors for A with corresponding eigenvalues λ_i .

Z

$$c(\lambda) = \det \begin{bmatrix} 1 - \lambda & 0 \\ 0 & -1 - \lambda \end{bmatrix} = (1 - \lambda)(-1 - \lambda) = \lambda^2 - 1 = 0,$$

so the eigenvalues are $\lambda = +1$ and $\lambda = -1$.

For $\lambda = +1$:

$$Z - I = \begin{bmatrix} 0 & 0 \\ 0 & -2 \end{bmatrix}, \quad (Z - I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ -2b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies b = 0$$

The component a is unconstrained, and normalization $|a|^2 = 1$ gives (choosing $a = 1$)

$$|z_+\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = |0\rangle$$

For $\lambda = -1$:

$$Z + I = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}, \quad (Z + I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 2a \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies a = 0,$$

and normalization $|b|^2 = 1$ gives (choosing $b = 1$)

$$|z_-\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix} = |1\rangle$$

The eigenvectors are orthonormal, and the diagonal representation is

$$Z = |z_+\rangle\langle z_+| - |z_-\rangle\langle z_-| = |0\rangle\langle 0| - |1\rangle\langle 1|$$

X

$$c(\lambda) = \det \begin{bmatrix} -\lambda & 1 \\ 1 & -\lambda \end{bmatrix} = \lambda^2 - 1 = 0,$$

so the eigenvalues are $\lambda = +1$ and $\lambda = -1$.

For $\lambda = +1$:

$$X - I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \quad (X - I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -a + b \\ a - b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies b = a$$

Normalization $2|a|^2 = 1$ gives (choosing $a = 1/\sqrt{2}$)

$$|x_+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

For $\lambda = -1$:

$$X + I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad (X + I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a + b \\ a + b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies b = -a$$

Normalization $2|a|^2 = 1$ gives (choosing $a = 1/\sqrt{2}$)

$$|x_-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

The eigenvectors are orthogonal, $\langle x_+ | x_- \rangle = \frac{1}{2}(1 \cdot 1 + 1 \cdot (-1)) = 0$. The diagonal representation is

$$X = |x_+\rangle\langle x_+| - |x_-\rangle\langle x_-| = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Y

$$c(\lambda) = \det \begin{bmatrix} -\lambda & -i \\ i & -\lambda \end{bmatrix} = \lambda^2 - (-i)(i) = \lambda^2 - 1 = 0,$$

so the eigenvalues are $\lambda = +1$ and $\lambda = -1$.

For $\lambda = +1$:

$$Y - I = \begin{bmatrix} -1 & -i \\ i & -1 \end{bmatrix}, \quad (Y - I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} -a - ib \\ ia - b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies b = ia$$

The first row is consistent with this, since $-a - i(ia) = -a + a = 0$. Normalization $|a|^2 + |ia|^2 = 2|a|^2 = 1$ gives (choosing $a = 1/\sqrt{2}$)

$$|y_+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}$$

For $\lambda = -1$:

$$Y + I = \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix}, \quad (Y + I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a - ib \\ ia + b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \implies b = -ia$$

The first row is consistent with this, since $a - i(-ia) = a - a = 0$. Normalization $2|a|^2 = 1$ gives (choosing $a = 1/\sqrt{2}$)

$$|y_-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -i \end{bmatrix} = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

The eigenvectors are orthogonal, $\langle y_+ | y_- \rangle = \frac{1}{2}(1 \cdot 1 + (-i)(-i)) = \frac{1}{2}(1 - 1) = 0$. The diagonal representation is

$$Y = |y_+\rangle\langle y_+| - |y_-\rangle\langle y_-| = \frac{1}{2} \begin{bmatrix} 1 & -i \\ i & 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

In each case the two eigenvalues are distinct, so none of the eigenspaces is degenerate. With respect to its own orthonormal eigenvectors, each of X , Y , and Z has the diagonal matrix representation

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Exercise 2.17

Show that a normal matrix is Hermitian if and only if it has real eigenvalues.

Let A be a normal matrix. By the spectral decomposition (Theorem 2.1), A is diagonal with respect to some orthonormal basis $|i\rangle$ for the vector space, that is,

$$A = \sum_i \lambda_i |i\rangle\langle i|,$$

where λ_i are the eigenvalues of A and each $|i\rangle$ is an eigenvector of A with eigenvalue λ_i . Each $|i\rangle\langle i|$ is Hermitian, and the adjoint is anti-linear, $(\sum_i a_i A_i)^\dagger = \sum_i a_i^* A_i^\dagger$ (2.33). Therefore

$$A^\dagger = \sum_i \lambda_i^* |i\rangle\langle i|$$

Suppose A is Hermitian, $A^\dagger = A$. Acting on the eigenvector $|j\rangle$, using $\langle i|j\rangle = \delta_{ij}$,

$$\lambda_j |j\rangle = A|j\rangle = A^\dagger |j\rangle = \lambda_j^* |j\rangle$$

Since $|j\rangle \neq 0$, $\lambda_j = \lambda_j^*$, so every eigenvalue of A is real.

Conversely, suppose every eigenvalue is real, $\lambda_i^* = \lambda_i$. Then

$$A^\dagger = \sum_i \lambda_i^* |i\rangle\langle i| = \sum_i \lambda_i |i\rangle\langle i| = A,$$

so A is Hermitian.

Exercise 2.19: Pauli matrices: Hermitian and unitary

Show that the Pauli matrices are Hermitian and unitary.

In a matrix representation the Hermitian conjugation operation takes the matrix of A to the conjugate-transpose matrix, $A^\dagger \equiv (A^*)^T$. An operator is Hermitian if $A^\dagger = A$, and a matrix U is unitary if $U^\dagger U = I$.

Hermitian.

$$X^\dagger = (X^*)^T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = X, \quad Z^\dagger = (Z^*)^T = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z,$$

$$Y^\dagger = (Y^*)^T = \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}^T = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = Y$$

Unitary. Since $X^\dagger = X$, $Y^\dagger = Y$ and $Z^\dagger = Z$, we have $P^\dagger P = P^2$ for each Pauli matrix P :

$$X^\dagger X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I,$$

$$Y^\dagger Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} (-i)(i) & 0 \\ 0 & (i)(-i) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I,$$

$$Z^\dagger Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Hence X , Y and Z are both Hermitian and unitary.

Exercise 2.22

Prove that two eigenvectors of a Hermitian operator with different eigenvalues are necessarily orthogonal.

Let A be a Hermitian operator, and let $|v_1\rangle$ and $|v_2\rangle$ be eigenvectors with

$$A|v_1\rangle = \lambda_1|v_1\rangle, \quad A|v_2\rangle = \lambda_2|v_2\rangle, \quad \lambda_1 \neq \lambda_2$$

A Hermitian operator is normal, so by Exercise 2.17 its eigenvalues are real, $\lambda_1^* = \lambda_1$.

The adjoint is defined by $(|v\rangle, A|w\rangle) = (A^\dagger|v\rangle, |w\rangle)$ (2.32), and the inner product is linear in the second argument and conjugate-linear in the first. Consider $(|v_1\rangle, A|v_2\rangle)$. Using linearity in the second argument,

$$(|v_1\rangle, A|v_2\rangle) = \lambda_2 \langle v_1|v_2\rangle$$

Using $A^\dagger = A$ and conjugate-linearity in the first argument,

$$(|v_1\rangle, A|v_2\rangle) = (A^\dagger|v_1\rangle, |v_2\rangle) = (A|v_1\rangle, |v_2\rangle) = (\lambda_1|v_1\rangle, |v_2\rangle) = \lambda_1^* \langle v_1|v_2\rangle = \lambda_1 \langle v_1|v_2\rangle$$

Equating the two expressions,

$$(\lambda_1 - \lambda_2)\langle v_1 | v_2 \rangle = 0$$

Since $\lambda_1 \neq \lambda_2$, it follows that $\langle v_1 | v_2 \rangle = 0$, so $|v_1\rangle$ and $|v_2\rangle$ are orthogonal.

Exercise 2.26

Let $|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Write out $|\psi\rangle^{\otimes 2}$ and $|\psi\rangle^{\otimes 3}$ explicitly, both in terms of tensor products like $|0\rangle|1\rangle$ and using the Kronecker product.

Tensor products.

$$|\psi\rangle^{\otimes 2} = |\psi\rangle \otimes |\psi\rangle = \frac{1}{2}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle) = \frac{1}{2}(|0\rangle|0\rangle + |0\rangle|1\rangle + |1\rangle|0\rangle + |1\rangle|1\rangle)$$

$$\begin{aligned} |\psi\rangle^{\otimes 3} &= |\psi\rangle \otimes |\psi\rangle^{\otimes 2} \\ &= \frac{1}{2\sqrt{2}}(|0\rangle|0\rangle|0\rangle + |0\rangle|0\rangle|1\rangle + |0\rangle|1\rangle|0\rangle + |0\rangle|1\rangle|1\rangle \\ &\quad + |1\rangle|0\rangle|0\rangle + |1\rangle|0\rangle|1\rangle + |1\rangle|1\rangle|0\rangle + |1\rangle|1\rangle|1\rangle) \end{aligned}$$

Kronecker product. With $|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$,

$$|\psi\rangle^{\otimes 2} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \times 1 \\ 1 \times 1 \\ 1 \times 1 \\ 1 \times 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

$$|\psi\rangle^{\otimes 3} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \otimes \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{2\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix},$$

Exercise 2.27

Calculate the matrix representation of the tensor products of the Pauli operators (a) X and Z ; (b) I and X ; (c) X and I . Is the tensor product commutative?

Using the Kronecker product (2.50):

(a)

$$X \otimes Z = \begin{bmatrix} 0 \cdot Z & 1 \cdot Z \\ 1 \cdot Z & 0 \cdot Z \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix}$$

(b)

$$I \otimes X = \begin{bmatrix} 1 \cdot X & 0 \cdot X \\ 0 \cdot X & 1 \cdot X \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

(c)

$$X \otimes I = \begin{bmatrix} 0 \cdot I & 1 \cdot I \\ 1 \cdot I & 0 \cdot I \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

The matrices in (b) and (c) are different, so $I \otimes X \neq X \otimes I$. The tensor product is not commutative.

Exercise 2.31

Show that the tensor product of two positive operators is positive.

Let A and B be positive operators on V and W . A positive operator is Hermitian, so by the spectral decomposition it has a diagonal representation with non-negative eigenvalues:

$$A = \sum_i a_i |a_i\rangle\langle a_i|, \quad B = \sum_j b_j |b_j\rangle\langle b_j|, \quad a_i \geq 0, \quad b_j \geq 0,$$

where $|a_i\rangle$ and $|b_j\rangle$ are orthonormal bases for V and W . By the definition (2.45) of the tensor product of operators,

$$A \otimes B = \sum_{i,j} a_i b_j |a_i\rangle\langle a_i| \otimes |b_j\rangle\langle b_j| = \sum_{i,j} a_i b_j (|a_i\rangle \otimes |b_j\rangle) (\langle a_i| \otimes \langle b_j|)$$

The vectors $|a_i\rangle \otimes |b_j\rangle$ form a basis for $V \otimes W$, and by the inner product (2.49) they are orthonormal. Let $|\varphi\rangle$ be any vector in $V \otimes W$, and write $|\varphi\rangle = \sum_{i,j} c_{ij} |a_i\rangle \otimes |b_j\rangle$. Then

$$\langle \varphi | A \otimes B | \varphi \rangle = \sum_{i,j} a_i b_j |\langle a_i | \langle b_j | \varphi \rangle|^2 = \sum_{i,j} a_i b_j |c_{ij}|^2 \geq 0,$$

since each $a_i b_j \geq 0$. The quantity $(|\varphi\rangle, A \otimes B |\varphi\rangle)$ is therefore real and non-negative for every $|\varphi\rangle$, so $A \otimes B$ is positive.

Exercise 2.34

Find the square root and logarithm of the matrix

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix}$$

A is Hermitian, so it has a spectral decomposition $A = \sum_a a |a\rangle\langle a|$, and a function of A is defined by $f(A) \equiv \sum_a f(a) |a\rangle\langle a|$.

The characteristic equation is

$$c(\lambda) = \det |A - \lambda I| = (4 - \lambda)^2 - 9 = \lambda^2 - 8\lambda + 7 = (\lambda - 7)(\lambda - 1) = 0,$$

so the eigenvalues are 7 and 1. The normalized eigenvectors are

$$|a_7\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |a_1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

which satisfy $A|a_7\rangle = 7|a_7\rangle$ and $A|a_1\rangle = |a_1\rangle$. The diagonal representation is

$$A = 7|a_7\rangle\langle a_7| + |a_1\rangle\langle a_1| = \frac{7}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Square root.

$$\sqrt{A} = \sqrt{7}|a_7\rangle\langle a_7| + |a_1\rangle\langle a_1| = \frac{1}{2} \begin{bmatrix} \sqrt{7} + 1 & \sqrt{7} - 1 \\ \sqrt{7} - 1 & \sqrt{7} + 1 \end{bmatrix}$$

Logarithm.

$$\log A = \log 7 |a_7\rangle\langle a_7| + \log 1 |a_1\rangle\langle a_1| = \frac{\log 7}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Exercise 2.35: Exponential of the Pauli matrices

Let $\vec{v}$ be any real, three-dimensional unit vector and θ a real number. Prove that

$$\exp(i\theta \vec{v} \cdot \vec{\sigma}) = \cos(\theta)I + i \sin(\theta) \vec{v} \cdot \vec{\sigma}, \quad \vec{v} \cdot \vec{\sigma} \equiv \sum_{i=1}^3 v_i \sigma_i$$

With $\sigma_1 = X$, $\sigma_2 = Y$, $\sigma_3 = Z$,

$$\vec{v} \cdot \vec{\sigma} = \begin{bmatrix} v_3 & v_1 - iv_2 \\ v_1 + iv_2 & -v_3 \end{bmatrix}$$

Since the v_i are real and each Pauli matrix is Hermitian, $\vec{v} \cdot \vec{\sigma}$ is Hermitian. Its characteristic equation is

$$c(\lambda) = \det |\vec{v} \cdot \vec{\sigma} - \lambda I| = (v_3 - \lambda)(-v_3 - \lambda) - (v_1 - iv_2)(v_1 + iv_2) = \lambda^2 - (v_1^2 + v_2^2 + v_3^2) = \lambda^2 - 1,$$

using $|\vec{v}| = 1$. The eigenvalues are +1 and -1. Let $|e_+\rangle$ and $|e_-\rangle$ be the corresponding normalized eigenvectors. They are orthogonal, since they are eigenvectors of a Hermitian operator with different eigenvalues (Exercise 2.22), so $|e_+\rangle$, $|e_-\rangle$ is an orthonormal basis. The spectral decomposition and the completeness relation give

$$\vec{v} \cdot \vec{\sigma} = |e_+\rangle\langle e_+| - |e_-\rangle\langle e_-|, \quad I = |e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|$$

By the definition of a function of an operator,

$$\exp(i\theta \vec{v} \cdot \vec{\sigma}) = e^{i\theta}|e_+\rangle\langle e_+| + e^{-i\theta}|e_-\rangle\langle e_-|$$

Substituting $e^{\pm i\theta} = \cos \theta \pm i \sin \theta$,

$$\exp(i\theta \vec{v} \cdot \vec{\sigma}) = \cos \theta (|e_+\rangle\langle e_+| + |e_-\rangle\langle e_-|) + i \sin \theta (|e_+\rangle\langle e_+| - |e_-\rangle\langle e_-|) = \cos(\theta)I + i \sin(\theta) \vec{v} \cdot \vec{\sigma}$$

Exercise 2.37: Cyclic property of the trace

If A and B are two linear operators, show that $\text{tr}(AB) = \text{tr}(BA)$.

The trace of an operator is the trace of any matrix representation of it, $\text{tr}(A) \equiv \sum_i A_{ii}$. Take matrix representations of A and B with respect to the same orthonormal basis; the matrix representation of AB is then the matrix product. Then

$$\text{tr}(AB) = \sum_i (AB)_{ii} = \sum_i \sum_j A_{ij} B_{ji} = \sum_j \sum_i B_{ji} A_{ij} = \sum_j (BA)_{jj} = \text{tr}(BA)$$

Exercise 2.39: The Hilbert-Schmidt inner product on operators

Let L_V be the vector space of linear operators on a Hilbert space V of dimension d .

(1) Show that $(A, B) \equiv \text{tr}(A^\dagger B)$ is an inner product function on $L_V \times L_V$.

The three requirements of an inner product are checked in turn. Matrix representations are taken with respect to a fixed orthonormal basis.

Linearity in the second argument. By the linearity of the trace,

$$\left(A, \sum_i \lambda_i B_i\right) = \text{tr}\left(A^\dagger \sum_i \lambda_i B_i\right) = \sum_i \lambda_i \text{tr}(A^\dagger B_i) = \sum_i \lambda_i (A, B_i)$$

Conjugate symmetry. For any matrix M , $(M^\dagger)_{ii} = M_{ii}^*$, so $\text{tr}(M)^* = \text{tr}(M^\dagger)$. Hence

$$(A, B)^* = \text{tr}(A^\dagger B)^* = \text{tr}((A^\dagger B)^\dagger) = \text{tr}(B^\dagger A) = (B, A)$$

Positivity.

$$(A, A) = \text{tr}(A^\dagger A) = \sum_{i,j} (A^\dagger)_{ij} A_{ji} = \sum_{i,j} A_{ji}^* A_{ji} = \sum_{i,j} |A_{ji}|^2 \geq 0,$$

with equality if and only if every matrix element A_{ji} is zero, that is, if and only if $A = 0$.

(2) Show that L_V has dimension d^2 .

Let $|1\rangle, \dots, |d\rangle$ be an orthonormal basis for V and define $E_{ij} \equiv |i\rangle\langle j|$. By the outer product representation (2.24), any operator can be written

$$A = \sum_{i,j} \langle i|A|j\rangle |i\rangle\langle j| = \sum_{i,j} \langle i|A|j\rangle E_{ij},$$

so the d^2 operators E_{ij} span L_V . They are linearly independent: if $\sum_{i,j} c_{ij} E_{ij} = 0$, then applying $\langle k | \cdot | l \rangle$ gives $c_{kl} = 0$ for all k, l . Hence $\{E_{ij}\}$ is a basis and L_V has dimension d^2 . Moreover,

$$(E_{ij}, E_{kl}) = \text{tr}(|j\rangle\langle i|k\rangle\langle l|) = \delta_{ik} \text{tr}(|j\rangle\langle l|) = \delta_{ik}\delta_{jl},$$

so the E_{ij} are orthonormal with respect to the Hilbert-Schmidt inner product.

(3) Find an orthonormal basis of Hermitian matrices for L_V .

Since $E_{ij}^\dagger = E_{ji}$, the following d^2 operators are Hermitian:

$$E_{ii} \ (1 \leq i \leq d), \quad S_{ij} = \frac{E_{ij} + E_{ji}}{\sqrt{2}}, \quad T_{ij} = \frac{i(E_{ij} - E_{ji})}{\sqrt{2}} \quad (1 \leq i < j \leq d)$$

Their orthonormality follows from $(E_{ij}, E_{kl}) = \delta_{ik}\delta_{jl}$. For $i < j$ and $k < l$ the terms $\delta_{il}\delta_{jk}$ vanish, so

$$(S_{ij}, S_{kl}) = \frac{1}{2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk} + \delta_{jk}\delta_{il} + \delta_{jl}\delta_{ik}) = \delta_{ik}\delta_{jl},$$

$$(T_{ij}, T_{kl}) = \frac{1}{2}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk} - \delta_{jk}\delta_{il} + \delta_{jl}\delta_{ik}) = \delta_{ik}\delta_{jl},$$

$$(S_{ij}, T_{kl}) = \frac{i}{2}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk} + \delta_{jk}\delta_{il} - \delta_{jl}\delta_{ik}) = 0,$$

and $(E_{mm}, S_{kl}) = (E_{mm}, T_{kl}) = 0$ for $k < l$, $(E_{mm}, E_{nn}) = \delta_{mn}$. There are $d + 2 \cdot \frac{d(d-1)}{2} = d^2$ of these operators, so by part (2) they form an orthonormal basis of Hermitian matrices for L_V .

Exercise 2.43

Show that for $j, k = 1, 2, 3$,

$$\sigma_j \sigma_k = \delta_{jk} I + i \sum_{l=1}^3 \epsilon_{jkl} \sigma_l,$$

where $\sigma_1 = X$, $\sigma_2 = Y$, $\sigma_3 = Z$, and $\epsilon_{jkl} = 0$ except for $\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1$ and $\epsilon_{321} = \epsilon_{213} = \epsilon_{132} = -1$.

Case $j = k$. Here $\delta_{jk} = 1$ and $\epsilon_{jkl} = 0$ for every l , so the right-hand side is I . From Exercise 2.19, $\sigma_j^2 = \sigma_j^\dagger \sigma_j = I$ for each Pauli matrix.

Case $j \neq k$. Here $\delta_{jk} = 0$, and the sum contains a single non-zero term, with l the index different from j and k . The six products are

$$XY = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} = iZ, \quad YX = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix} = -iZ,$$

$$YZ = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix} = iX, \quad ZY = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} = \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -iX,$$

$$ZX = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} = iY, \quad XZ = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -iY$$

The right-hand side in each case is

$$(j, k) = (1, 2) : i\epsilon_{123}\sigma_3 = iZ, \quad (2, 1) : i\epsilon_{213}\sigma_3 = -iZ,$$

$$(2, 3) : i\epsilon_{231}\sigma_1 = iX, \quad (3, 2) : i\epsilon_{321}\sigma_1 = -iX,$$

$$(3, 1) : i\epsilon_{312}\sigma_2 = iY, \quad (1, 3) : i\epsilon_{132}\sigma_2 = -iY,$$

which agrees with each product above. The identity therefore holds for all $j, k = 1, 2, 3$.

Exercise 2.47

Suppose A and B are Hermitian. Show that $i[A, B]$ is Hermitian.

Using $(AB)^\dagger = B^\dagger A^\dagger$ and the anti-linearity of the adjoint (2.33),

$$[A, B]^\dagger = (AB - BA)^\dagger = B^\dagger A^\dagger - A^\dagger B^\dagger = [B^\dagger, A^\dagger]$$

Since $A^\dagger = A$ and $B^\dagger = B$,

$$[A, B]^\dagger = [B, A] = BA - AB = -[A, B]$$

Therefore

$$(i[A, B])^\dagger = i^* [A, B]^\dagger = (-i)(-[A, B]) = i[A, B],$$

so $i[A, B]$ is Hermitian.

Exercise 2.50

Find the left and right polar decompositions of the matrix

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

By the polar decomposition (Theorem 2.3), $A = UJ = KU$ with U unitary and $J \equiv \sqrt{A^\dagger A}$, $K \equiv \sqrt{AA^\dagger}$ positive operators. Since $\det A = 1$, A is invertible, so J is invertible and $U = AJ^{-1}$ is uniquely determined.

Step 1: $A^\dagger A$ and $AA^\dagger$.

$$A^\dagger A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, \quad AA^\dagger = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$

Step 2: The spectral decomposition of $A^\dagger A$ and the operator J . Let $M = A^\dagger A$. The characteristic equation is

$$c(\lambda) = \det |M - \lambda I| = (2 - \lambda)(1 - \lambda) - 1 = \lambda^2 - 3\lambda + 1 = 0,$$

so the eigenvalues are

$$\lambda_{\pm} = \frac{3 \pm \sqrt{5}}{2}$$

Both are positive and distinct. Let $P_{\pm}$ be the projectors onto the corresponding eigenspaces. Then

$$M = \lambda_+ P_+ + \lambda_- P_-, \quad P_+ + P_- = I, \quad P_i P_j = \delta_{ij} P_i$$

Since $\left(\frac{\sqrt{5}+1}{2}\right)^2 = \frac{6+2\sqrt{5}}{4} = \lambda_+$, the square roots of the eigenvalues are

$$\sqrt{\lambda_{\pm}} = \frac{\sqrt{5} \pm 1}{2}, \quad \sqrt{\lambda_+} - \sqrt{\lambda_-} = 1, \quad \lambda_+ - \lambda_- = \sqrt{5},$$

and

$$J = \sqrt{M} = \sqrt{\lambda_+} P_+ + \sqrt{\lambda_-} P_-$$

From $M - \lambda_- I = (\lambda_+ - \lambda_-) P_+ = \sqrt{5} P_+$,

$$P_+ = \frac{M - \lambda_- I}{\sqrt{5}}$$

Using $P_- = I - P_+$ and $\sqrt{\lambda_+} - \sqrt{\lambda_-} = 1$,

$$J = \sqrt{\lambda_-} I + (\sqrt{\lambda_+} - \sqrt{\lambda_-}) P_+ = \left(\sqrt{\lambda_-} - \frac{\lambda_-}{\sqrt{5}} \right) I + \frac{M}{\sqrt{5}}$$

The coefficient of I is

$$\sqrt{\lambda_-} - \frac{\lambda_-}{\sqrt{5}} = \frac{\sqrt{5} - 1}{2} \cdot \frac{2\sqrt{5} - (\sqrt{5} - 1)}{2\sqrt{5}} = \frac{(\sqrt{5} - 1)(\sqrt{5} + 1)}{4\sqrt{5}} = \frac{1}{\sqrt{5}},$$

so

$$J = \frac{M + I}{\sqrt{5}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$$

Check: J has eigenvalues $\sqrt{\lambda_{\pm}} > 0$, so it is positive, and

$$J^2 = \frac{1}{5} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = A^\dagger A$$

Step 3: The unitary U and the left polar decomposition. The inverse of J is

$$J^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix}, \quad JJ^{-1} = \frac{1}{5} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Then

$$U = AJ^{-1} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 3 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}$$

U is unitary:

$$U^\dagger U = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = I$$

The left polar decomposition is

$$A = UJ = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Step 4: The operator K and the right polar decomposition. The matrix $N = AA^\dagger$ has trace 3 and determinant 1, so its characteristic equation is also $\lambda^2 - 3\lambda + 1 = 0$, with the same eigenvalues $\lambda_\pm$. The argument of Step 2 applies with N in place of M :

$$K = \sqrt{AA^\dagger} = \frac{N + I}{\sqrt{5}} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$

Check:

$$K^2 = \frac{1}{5} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 5 \\ 5 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} = AA^\dagger$$

Since A is invertible, the unitary in $A = KU$ is the same U , and

$$A = KU = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \cdot \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 5 & 0 \\ 5 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Result.

$$A = UJ = KU, \quad U = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix}, \quad J = \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}, \quad K = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$$